

Activity 6 Pre-Lab — Dehydration of Cyclohexanol

[Your name] & [partner]
[MM/DD/YY]

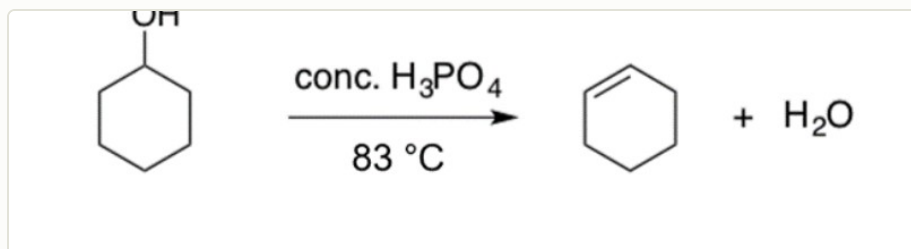
Organic Lab · Synthesis of cyclohexene + purification by simple distillation

■ **Black** = write this in your notebook ■ **Blue** = context, don't copy it Copy the header onto every page. Draw anything in a dashed box yourself.

Purpose

Synthesize **cyclohexene** by the acid-catalyzed **dehydration (E1)** of **cyclohexanol** using 85% phosphoric acid, isolate it by **simple distillation**, then check its purity/identity by TLC, the bromine test, and IR.

🔥 **DRAW** — reproduce this reaction in your notebook (from the procedure)



mechanism = **E1**: protonate the $-\text{OH}$ → lose H_2O (carbocation) → lose a $\beta\text{-H}$ → cyclohexene. Also write the conditions (85% H_3PO_4 , $\leq 83\text{ }^\circ\text{C}$, distill).

Chemical Reagents & Safety Table

CONTEXT ▶ Required: cyclohexanol, phosphoric acid, cyclohexene, sodium sulfate, sodium chloride. Typical SDS values — cross-check the SDS tile on Labflow.

Chemical	Role in lab	MW	Amount used	Physical properties	GHS pictograms
Cyclohexanol	Starting material (limiting)	100.16	10 mL (~9.6 g, ~0.096 mol)	Colorless liquid/waxy solid; bp $161\text{ }^\circ\text{C}$; d 0.96 g/mL; mp $-25\text{ }^\circ\text{C}$	 irritant
Phosphoric acid (85%)	Acid catalyst / dehydrating agent	98.00	12 mL	Colorless viscous liquid; d 1.69 g/mL	 corrosive
Cyclohexene	Product	82.14	product (theor. ~0.096 mol)	Colorless liquid; bp $83\text{ }^\circ\text{C}$; d 0.81 g/mL; flash pt $-12\text{ }^\circ\text{C}$	    flammable health irritant environ.
Sodium sulfate (anhydrous)	Drying agent	142.04	pea-sized	White solid; hygroscopic	— low hazard
Sodium chloride (sat. aq.)	Brine wash (removes water)	58.44	$2 \times 10\text{ mL}$	Colorless aqueous solution	— low hazard

FYI ▶ also used: 10% $\text{Na}_2\text{CO}_3(\text{aq})$ to neutralize phosphoric acid that distills over (fizzes → CO_2 , vent often!); charcoal boiling chips (regular stones dissolve in H_3PO_4).

GRAB FIRST: 100-mL round-bottom (reactor) · 50-mL round-bottom (receiver) · simple-distillation head + condenser + thermometer adapter + thermometer · heating mantle + Variac · lab jack + clamps · ice-water bath · small separatory funnel · 25-mL Erlenmeyer · charcoal boiling chips.

Procedure Plan (left = your plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Synthesis + distillation

1. Measure **~10 mL cyclohexanol** into the **100-mL RBF**; record exact volume.
2. Add **12 mL 85% H₃PO₄** + a few **charcoal boiling chips**, swirl to mix.
H₃PO₄ is corrosive — gloves; follow instructor's handling.
3. Clamp the reactor to the bars; build a **simple-distillation** apparatus (still head → thermometer → condenser → 50-mL receiver in an **ice-water bath**). Heating mantle on a **raised lab jack** below. *Jack must drop enough to slide the hot mantle out.*
4. Instructor checks setup → heat with the **Variac**.
5. Keep the **vapor temp ≤ 83 °C** (adjust Variac / lower mantle). Record the steady distilling temp. *83 °C = cyclohexene bp; leaves cyclohexanol + H₃PO₄ behind.*
6. Stop (lower jack, mantle off) when **5–10 mL** remain in the reactor. **Never distill to dryness.**
7. Neutralize: slowly add **10% Na₂CO₃(aq)** to the receiver in 5-mL portions, swirl till fizzing stops → transfer to sep funnel. *Vent often in the hood — CO₂ builds pressure.*
8. Wash the organic layer **2 × 10 mL saturated NaCl**; drain off the aqueous each time. *Product = organic (top) layer.*
9. Transfer product to a 25-mL Erlenmeyer; dry over a **pea-sized** amount of anhydrous **Na₂SO₄** until no longer cloudy.
10. Decant the clear cyclohexene into a pre-weighed flask; weigh → mass of product → **% yield**.

Tests (do all before discarding product)

11. **TLC** (30% EtOAc/hexanes) · **Bromine test** (Br₂ decolorizes → alkene) · **IR** (look for =C–H ~3050, C=C ~1650; loss of broad O–H).

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% yield = (actual mol cyclohexene ÷ theoretical mol) × 100

Experiment Notes (in lab)

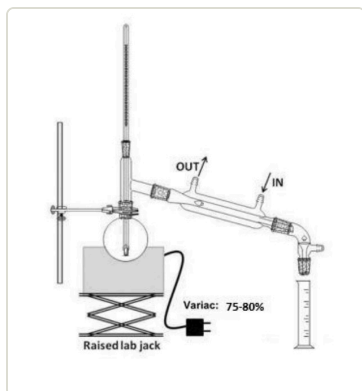
measurement	value
Exact cyclohexanol volume	mL
Steady vapor temp	°C
Mass empty flask	g
Mass flask + product	g
Mass cyclohexene	g
% yield	

Distillation / workup obs (layers, venting):

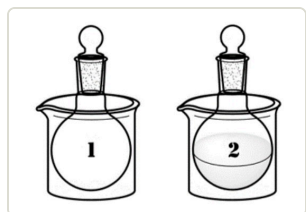
Product appearance (clear? turbid?):

Bromine test / TLC R_f / IR peaks:

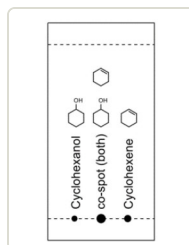
Setup & technique — draw these



simple distillation



weigh product (empty flask 1 → +product 2)



Part 2 TLC (3 lanes)

CONTEXT (don't copy) ▶

Why 83 °C cap? cyclohexene boils at 83 °C;

cyclohexanol (161 °C) & H₃PO₄ stay behind → pure

distillate.

Brine + Na_2SO_4 dry the product (cloudy → clear).

TLC lanes: cyclohexanol · co-spot · cyclohexene;

KMnO_4 -stain (cyclohexene is volatile — spot & develop fast).

Clean-up & Conclusions

- Acidic pot residue → neutralize/dispose per instructor; organic waste for cyclohexene after tests; aqueous washes → aqueous waste; return glassware; do your bench task.
- **Conclusions** (~4 sentences [after lab](#)): your % yield and a reason it wasn't 100% (product left in pot, evaporation, incomplete rxn); what the bromine test + IR $\text{C}=\text{C}$ confirmed; and how distillation purified the product.

Orgo Planner pre-lab guide · reproduce by hand in your numbered notebook · no names, safe to share.